Sorting: evaluation, improvements, and comparisons

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6 5 3 1 8 7 2 4

Overview

What is sorting

- Input
  - An array A of data records
  - A Key value for each data record
  - A comparison function
- Output
  - Permutation of A such that
  - if $i < j$, then $A[i] < A[j]$

Why sort

- Sorting algorithms are among the most frequently used algorithms in computer science

  Searching : Allows binary search of an N-element array in $O(\log N)$ time

  Top elements : Allows $O(1)$ time access to k-th largest element in the array for any k

  Detect duplicates : Allows easy detection of any duplicates

  ...

Evaluation of sorting algorithms

- Time

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  - some algorithms are good for *partially* sorted data
  - some are good if the keys are within a certain ranges. e.g., bucket sort.

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- Application dependency: e.g., we only want to have the top few.

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- External: data too large to fit in memory

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  - some are good if the keys are within a certain ranges. e.g., bucket sort.
- Application dependency: e.g., we only want to have the top few.
- External: data too large to fit in memory
- Parallel algorithms: data too large to run on one computer

Time: How fast is the algorithm

- The definition of a sorted array A says that for any $i < j$, $A[i] < A[j]$
- Need to at least check on each element at the very minimum, I.e., at least O(N)
- Could end up checking each element against every other element, which is $O(N^2)$.
- The big question is: How close to O(N) can you get?

Space

How much space does the sorting algorithm require in order to sort the collection of items?

- Is copying needed?
- In-place sorting: no copying. O(1) additional space
- some where in between for "temporary", e.g. $O(\log n)$ space
- External memory sorting: data so large that does not fit in memory

In-place algorithms

An algorithm is *in-place* if it uses only a small amount of memory in addition to that needed for the original input.

Whether the following algorithms are in-place sorting?

- Merge sort?

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- Merge sort?
- Insertion sort?

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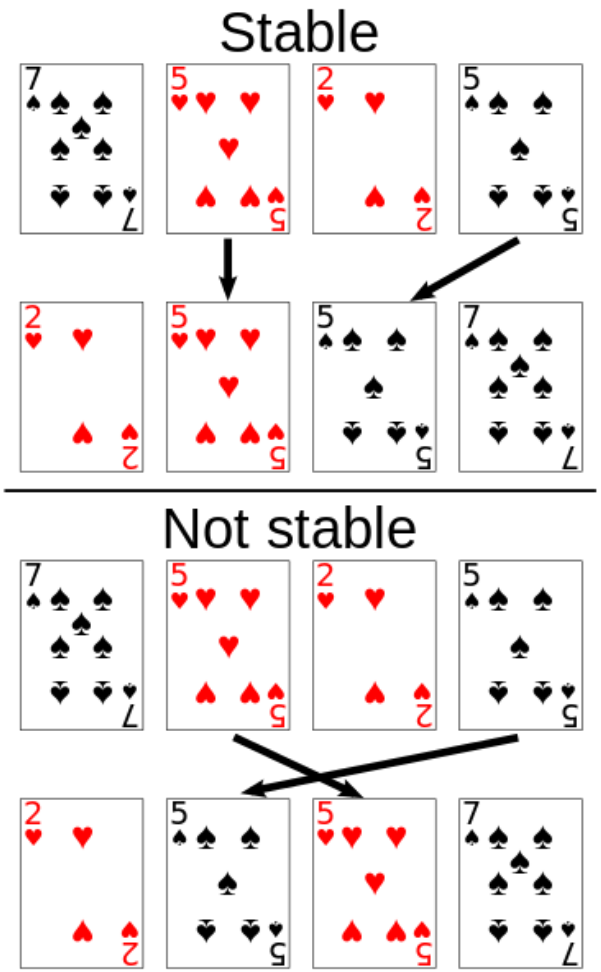
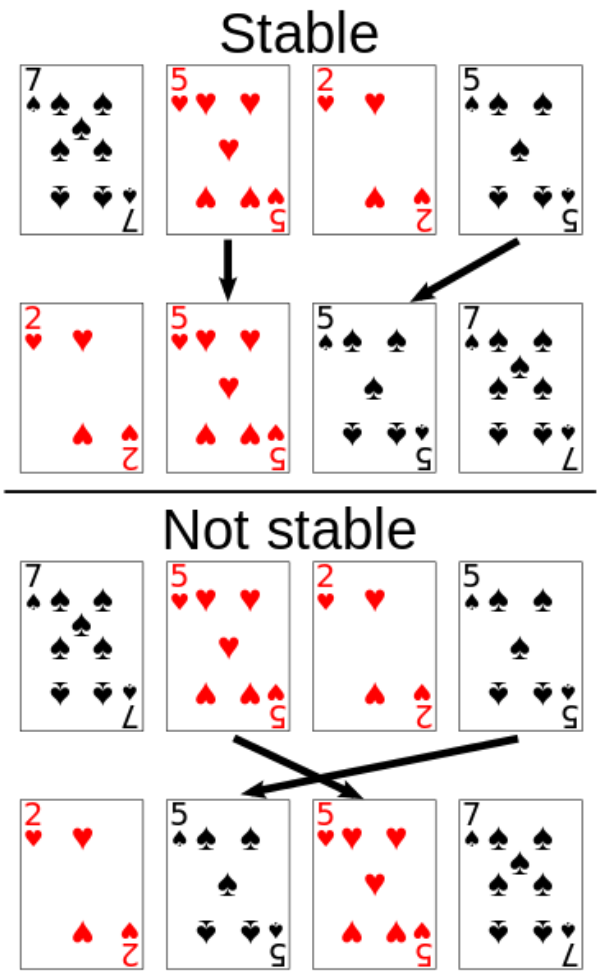


Does it rearrange the order of input data records which have the same key value (duplicates)?

- E.g. Phone book sorted by name. Now sort by county. Is the list still sorted by name within each county?
- Extremely important property for databases

Stability



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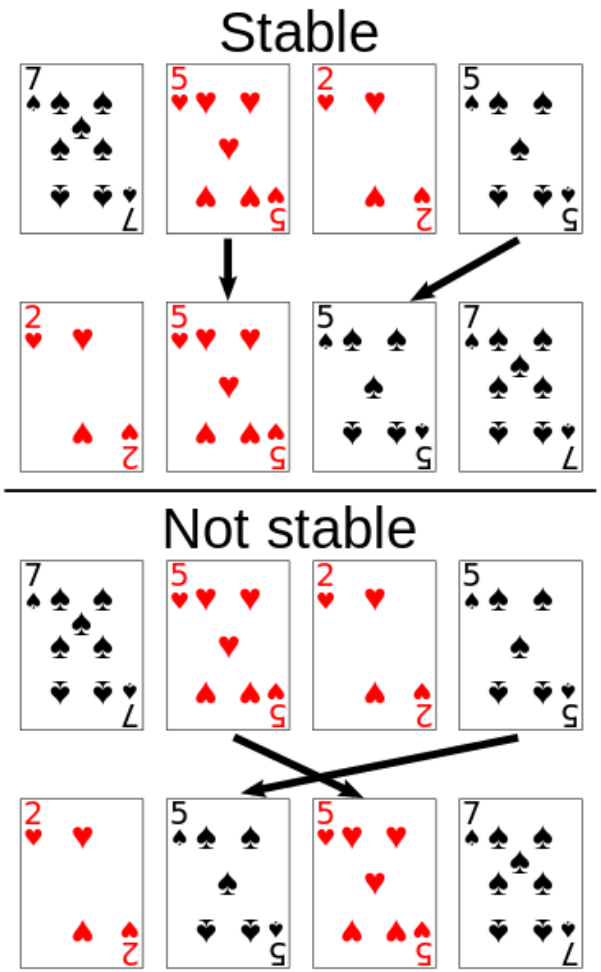

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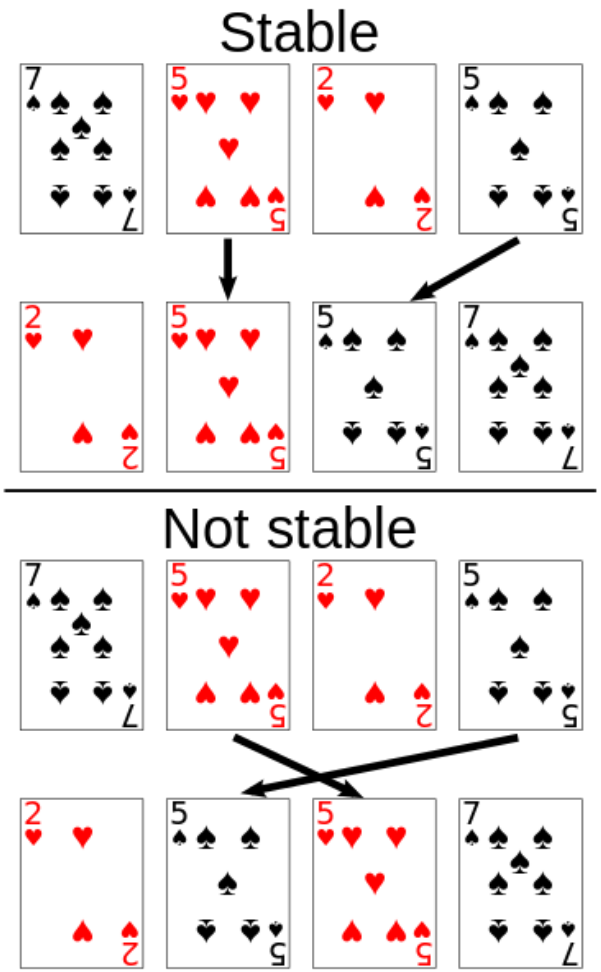

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- E.g. Phone book sorted by name. Now sort by county. Is the list still sorted by name within each county?
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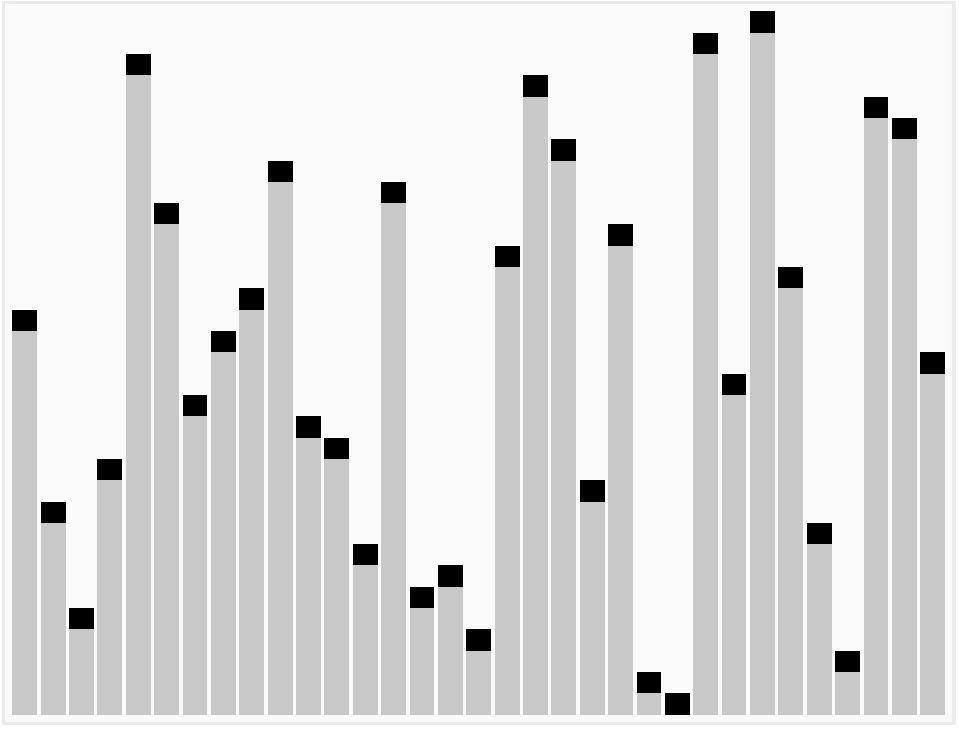
- Whether insertion sort is stable?
- How about merge sort?
- How about heap sort?

Quick sort



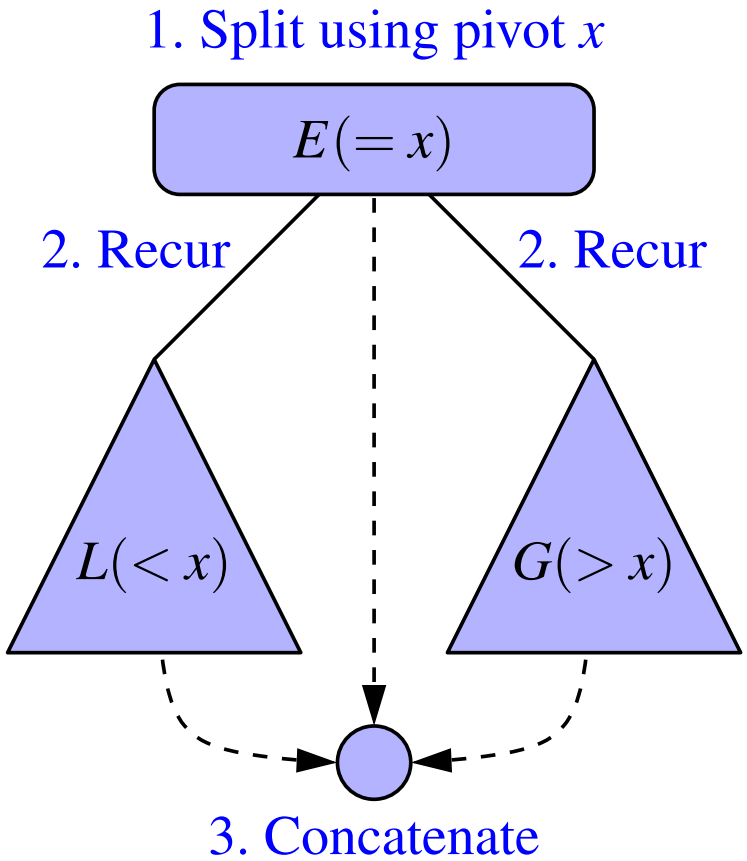
Overall idea: Divide-and Conquer

Quick-sort is a randomized sorting algorithm based on the divide-and-conquer paradigm

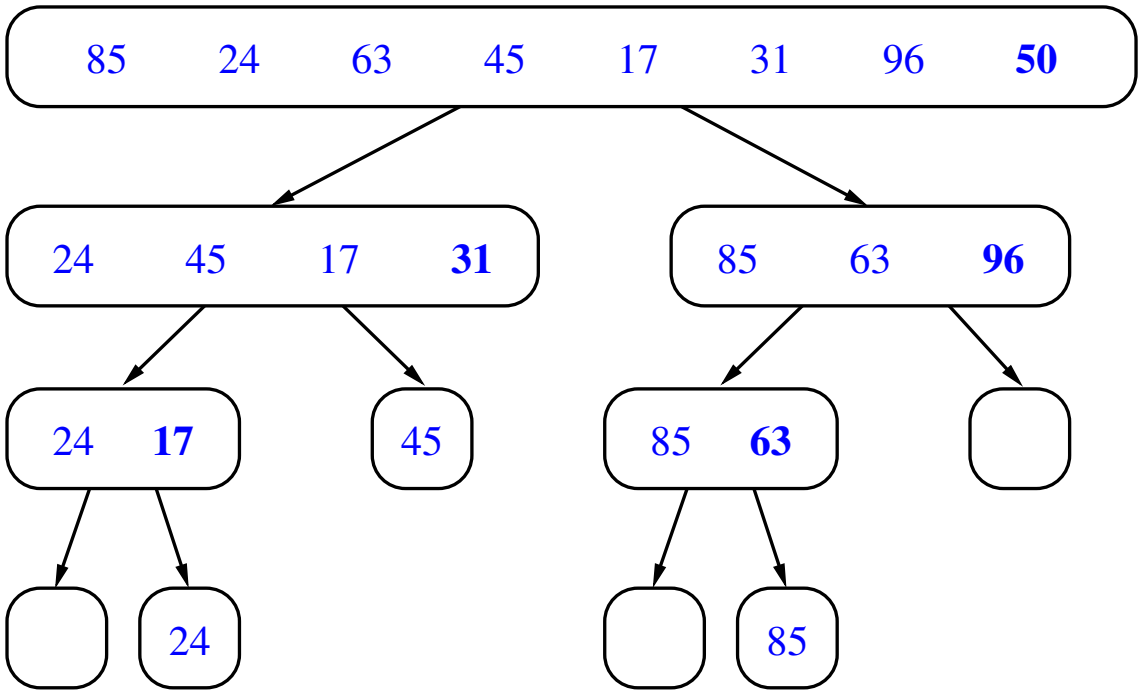
Divide pick a random element x (called pivot) and partition S into
- L elements less than x
- E elements equal x
- G elements greater than x

Conquer Recursively sort L and G

Combine join L, E and G

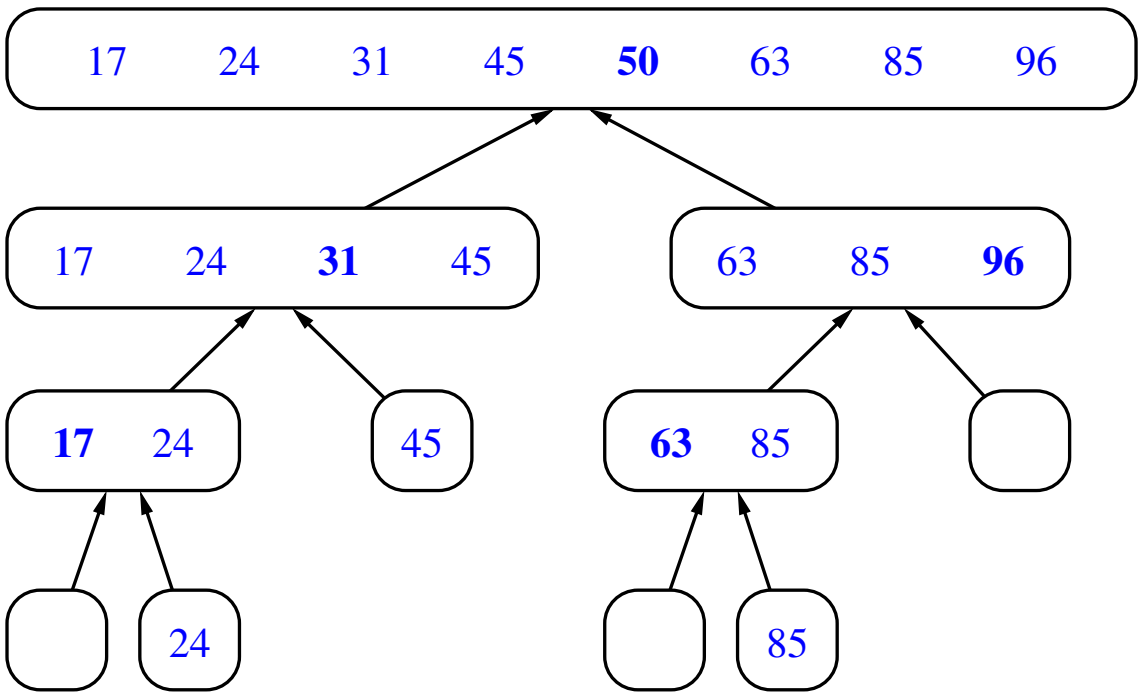


Quick sort example: divide



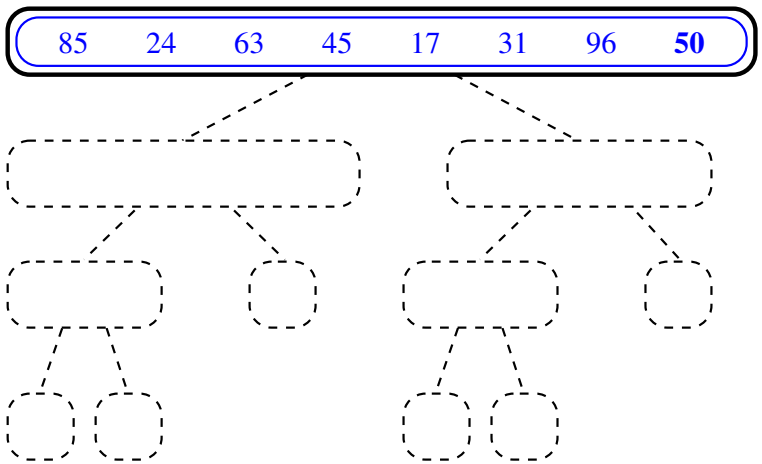
Divide process

Combine



Combination process

Quick sort step-by-step: recursive calls



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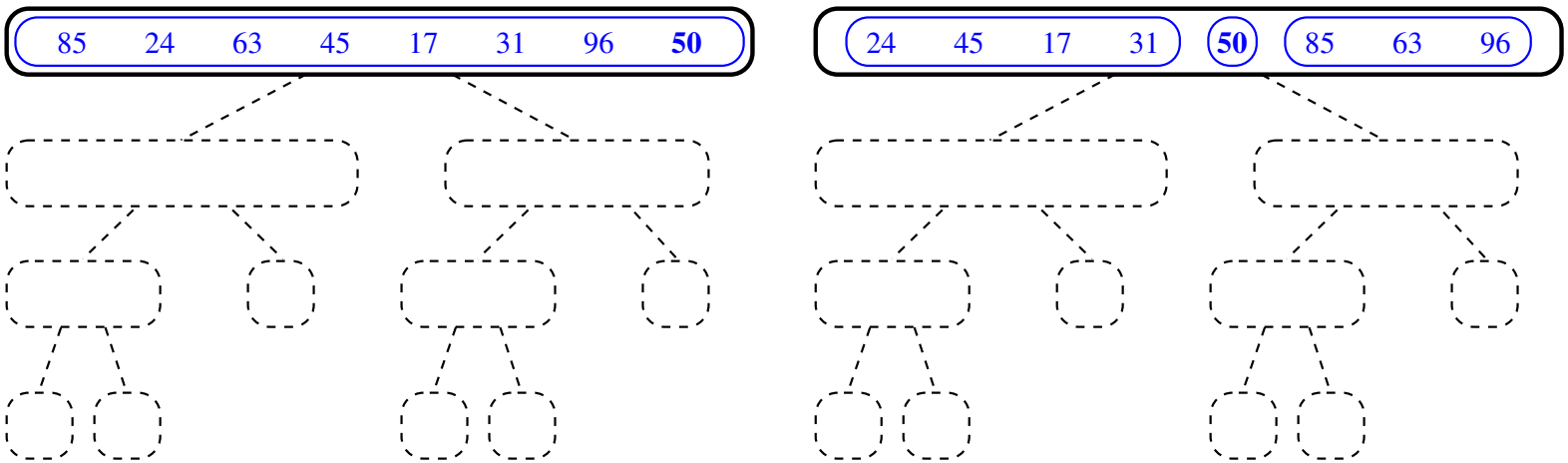

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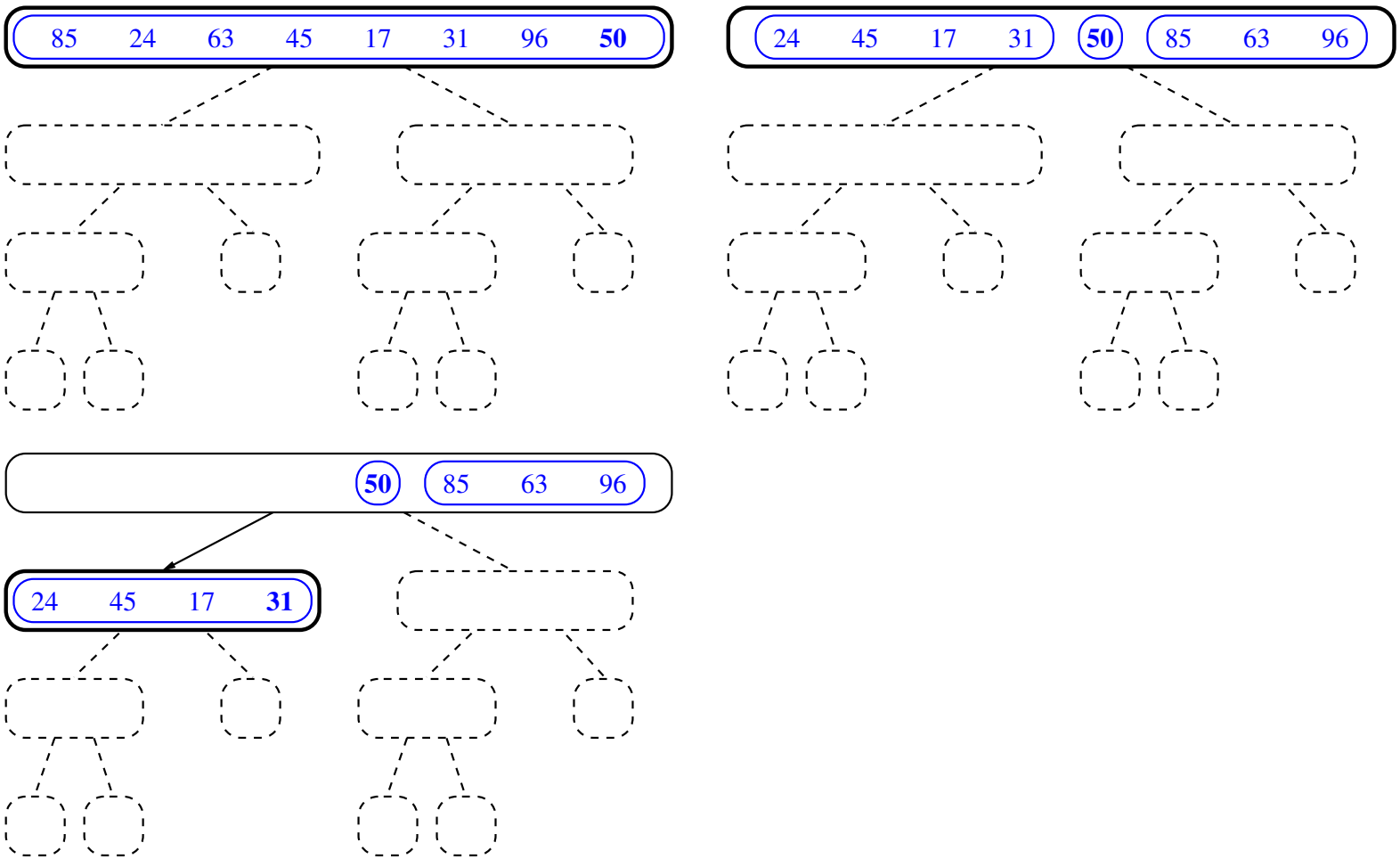

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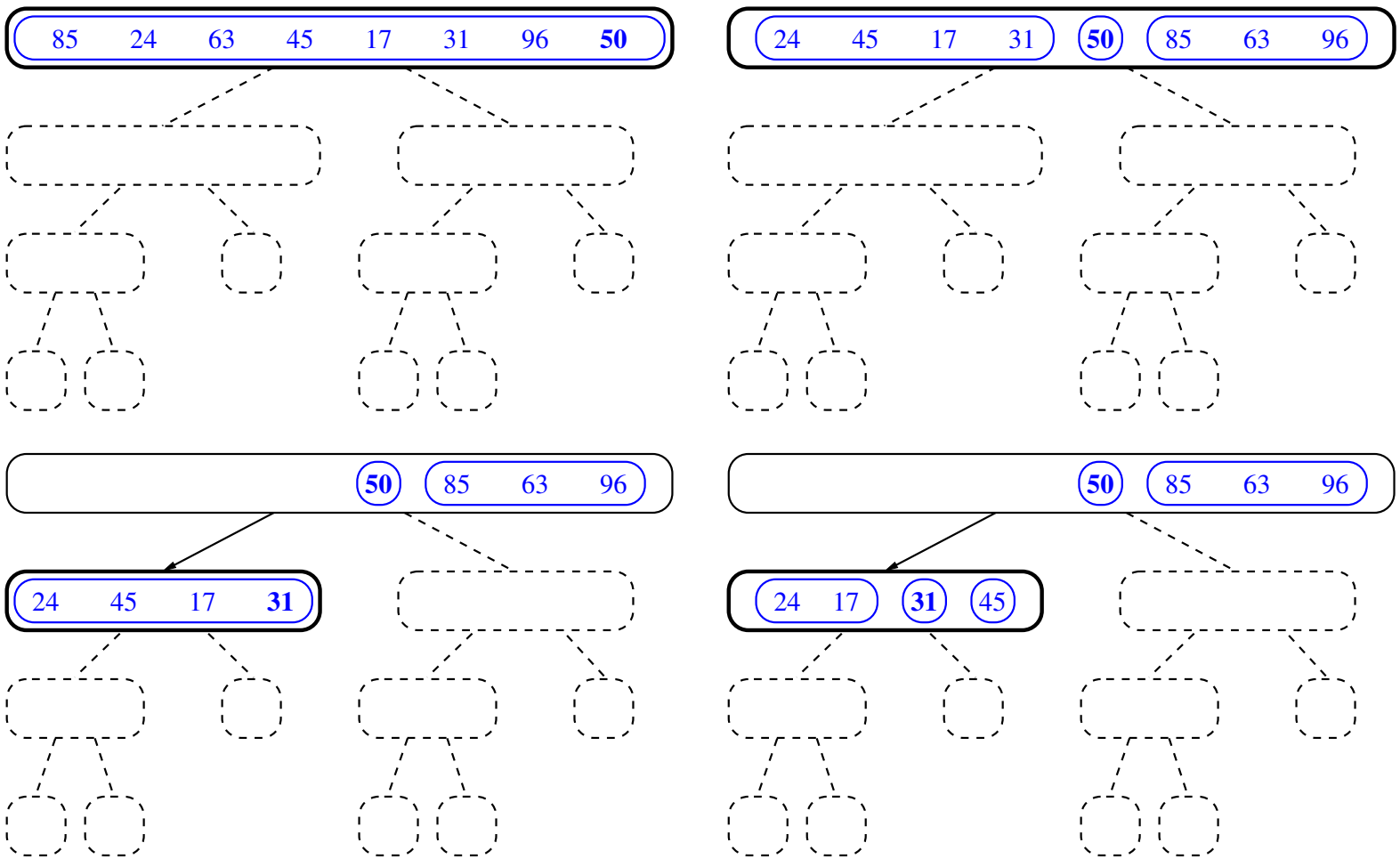
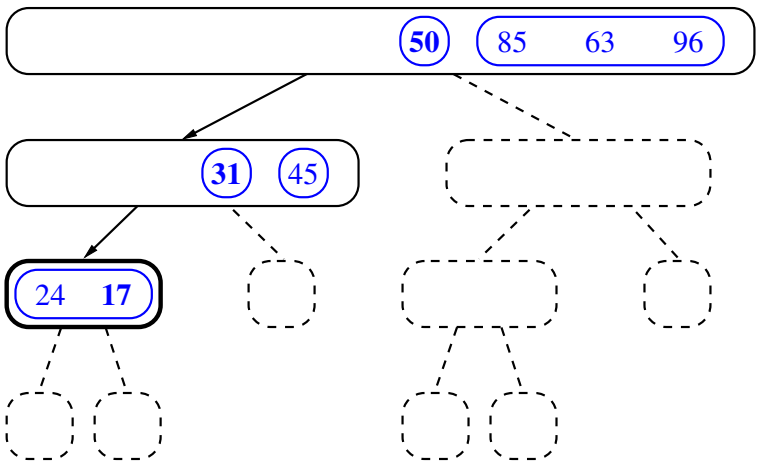


Recursive calls down to the base case



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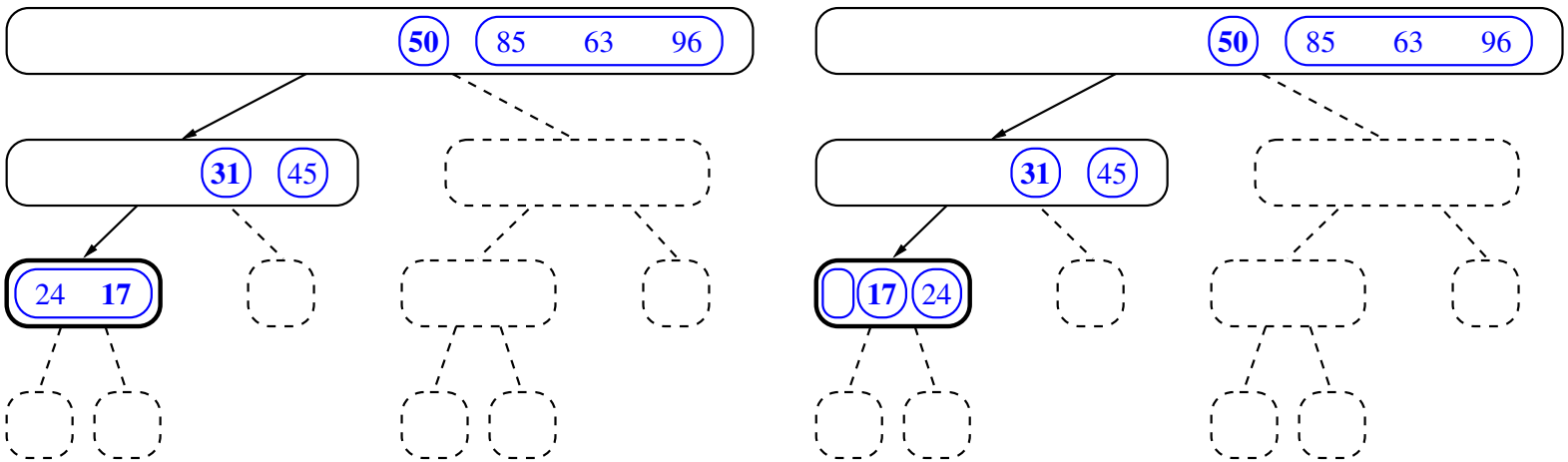

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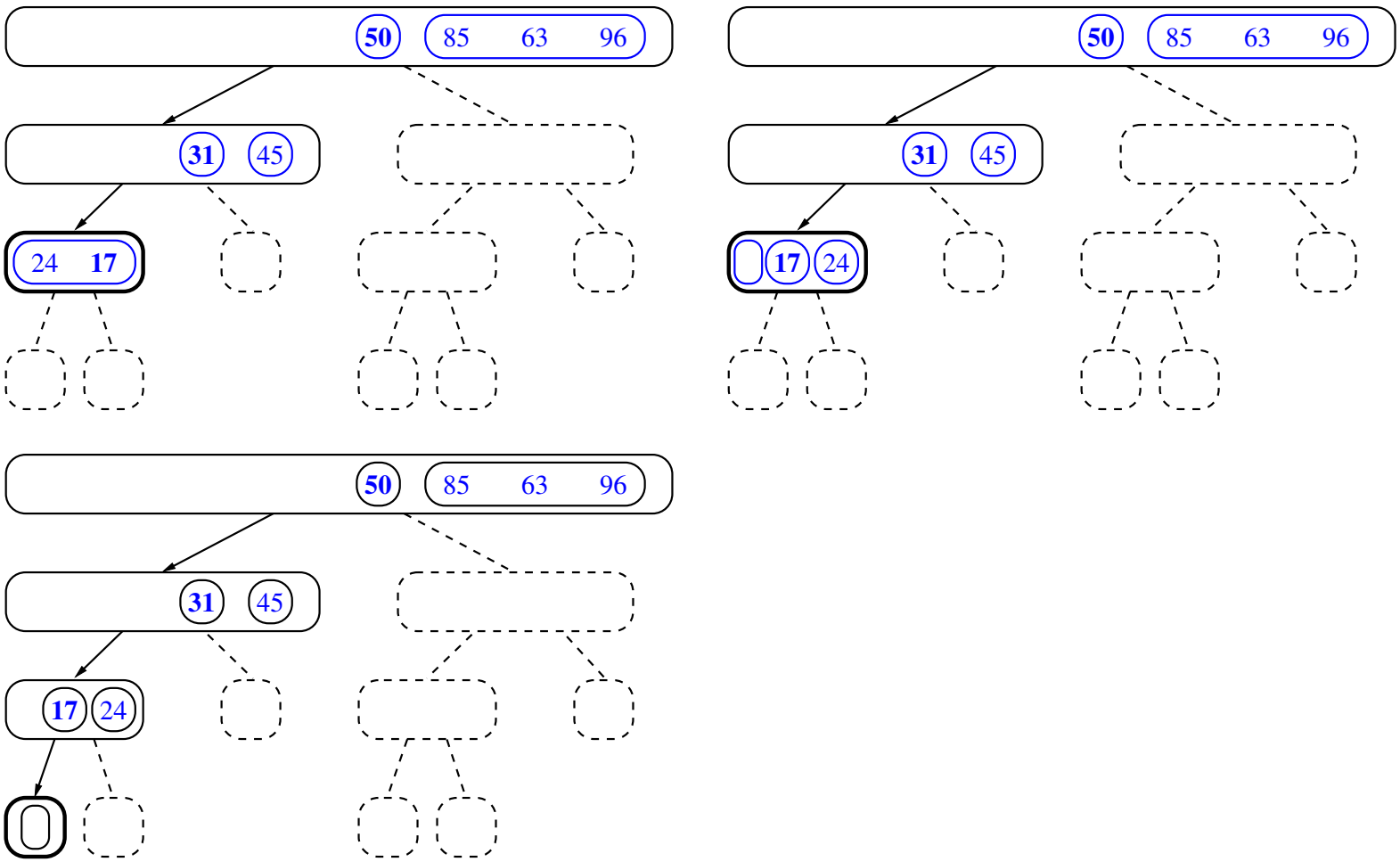

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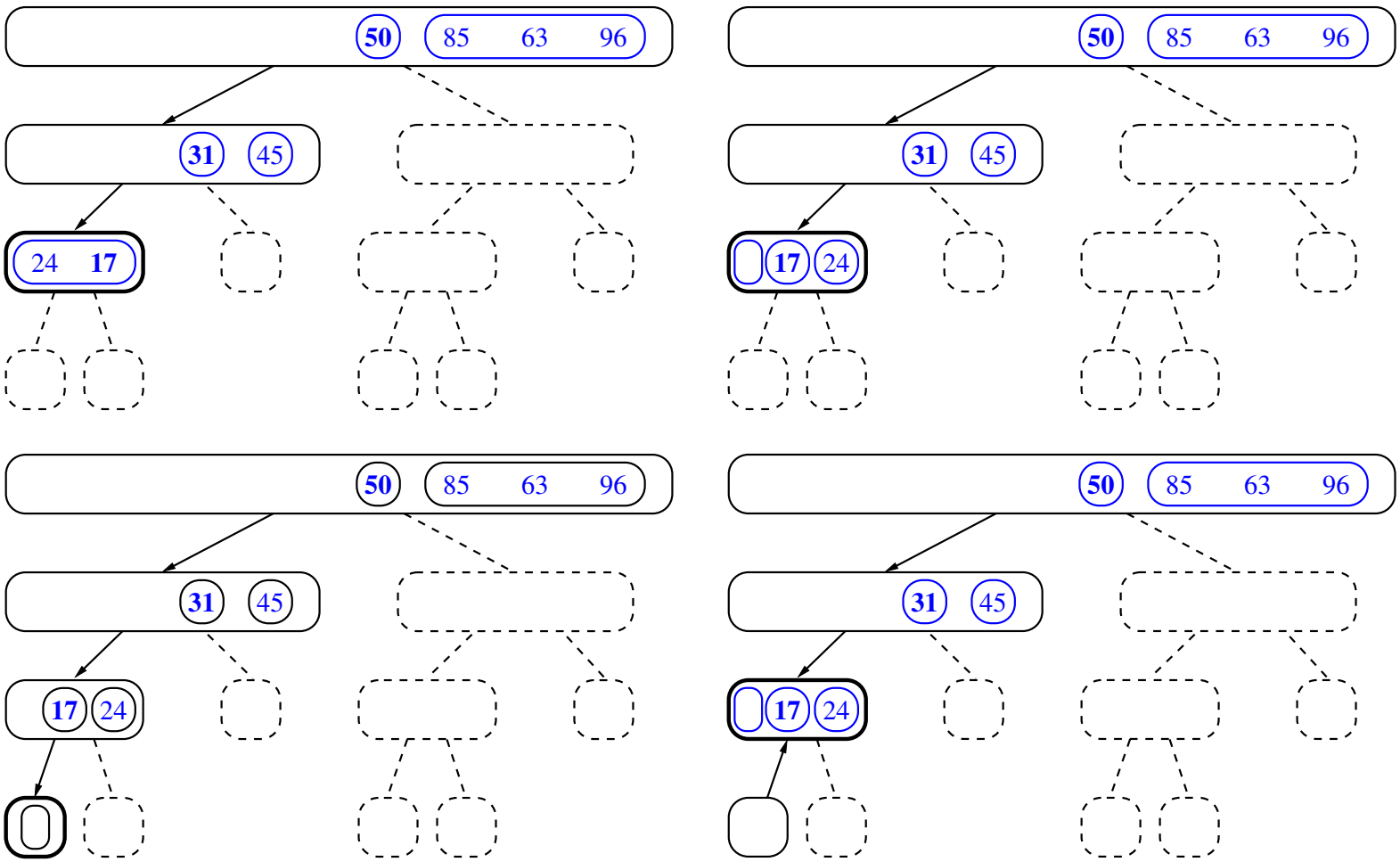
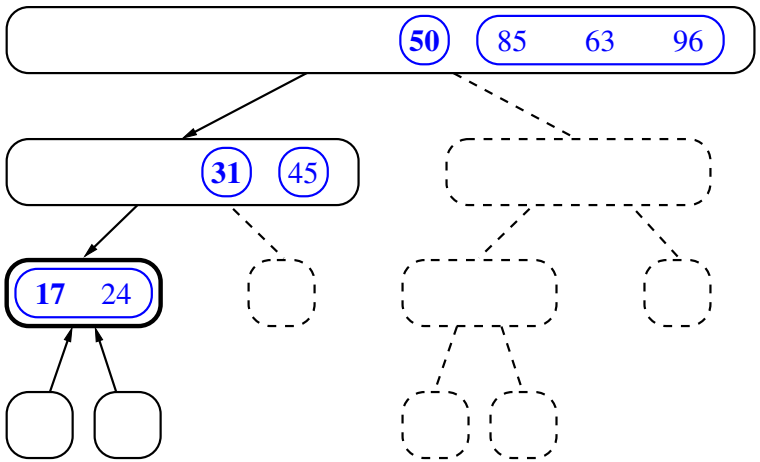


Return of recursive calls, then call (45) branch



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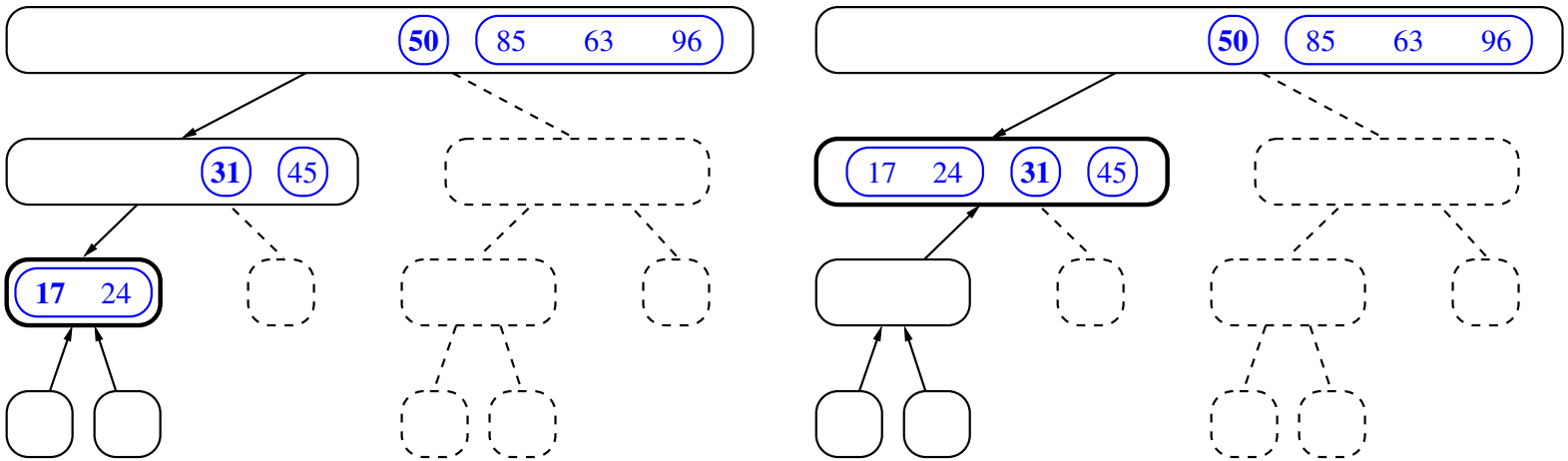

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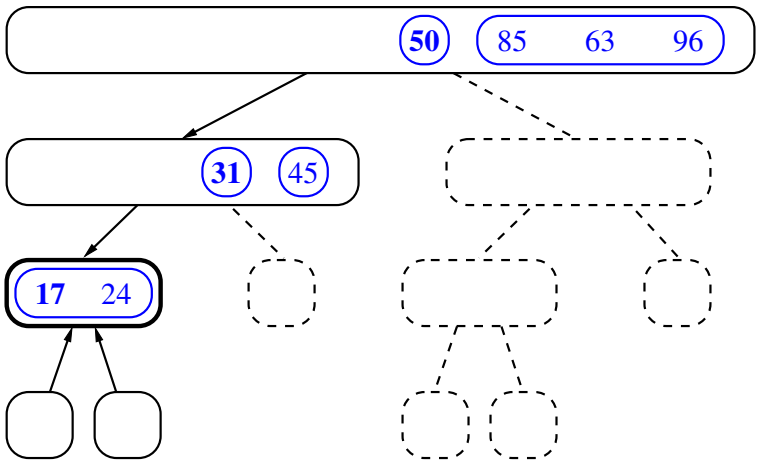

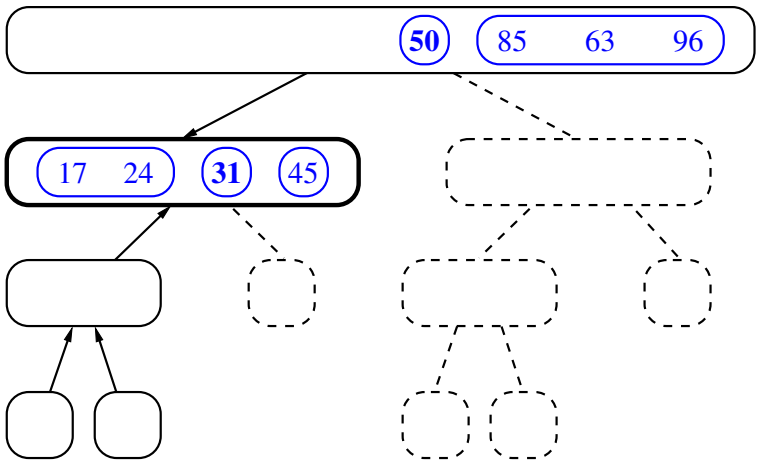

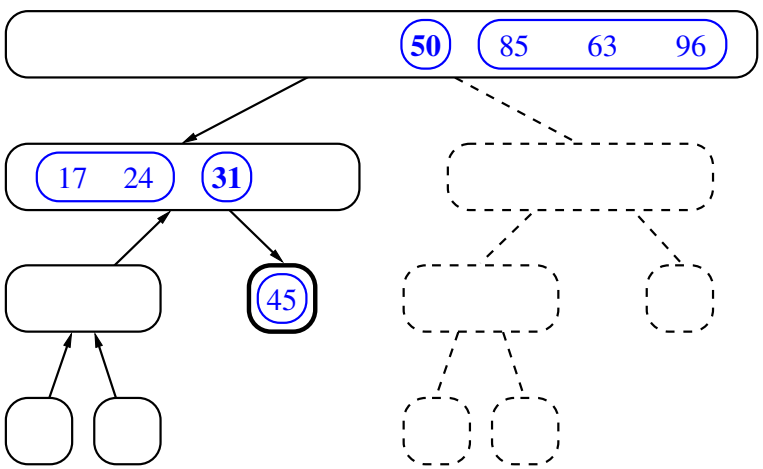

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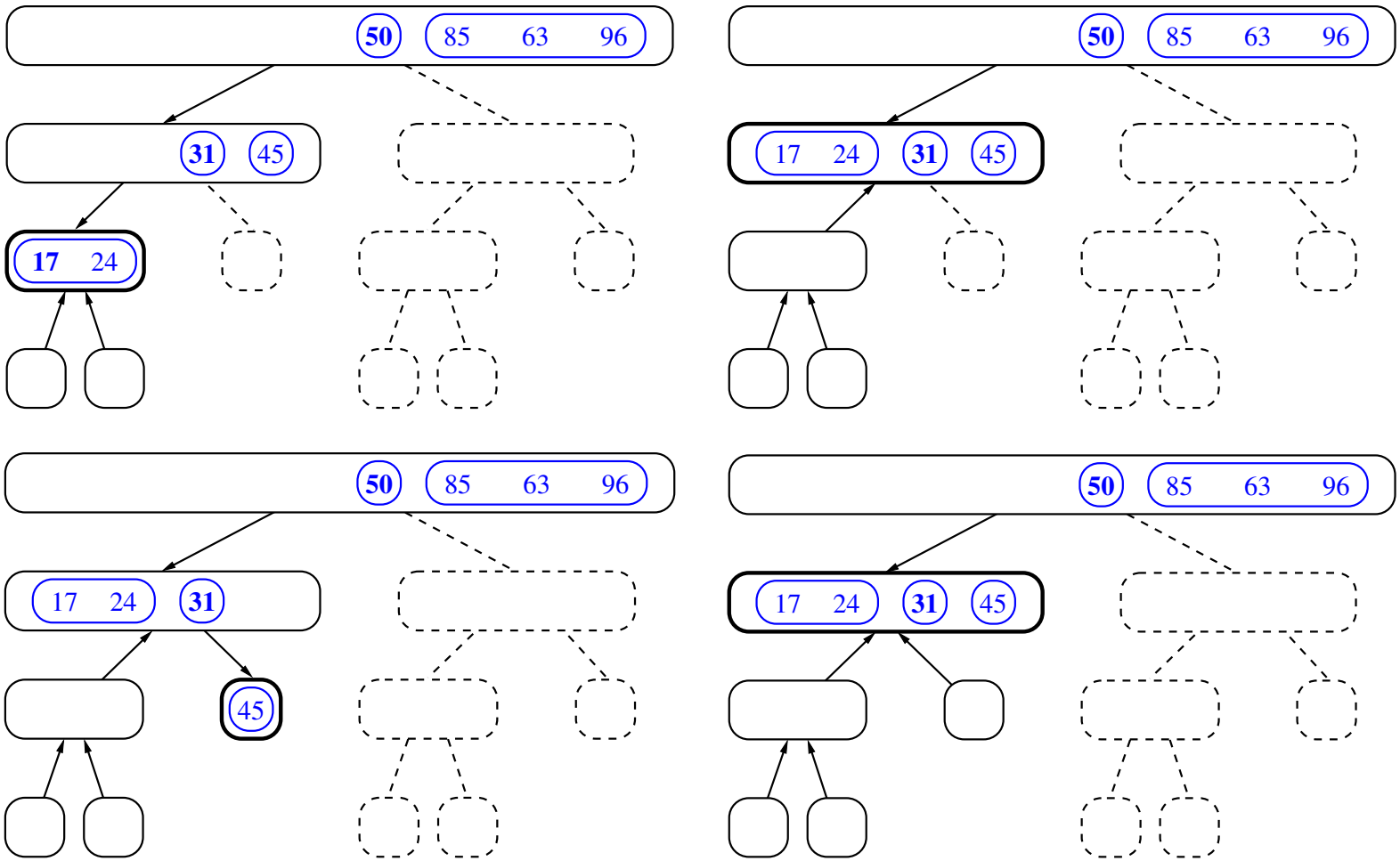

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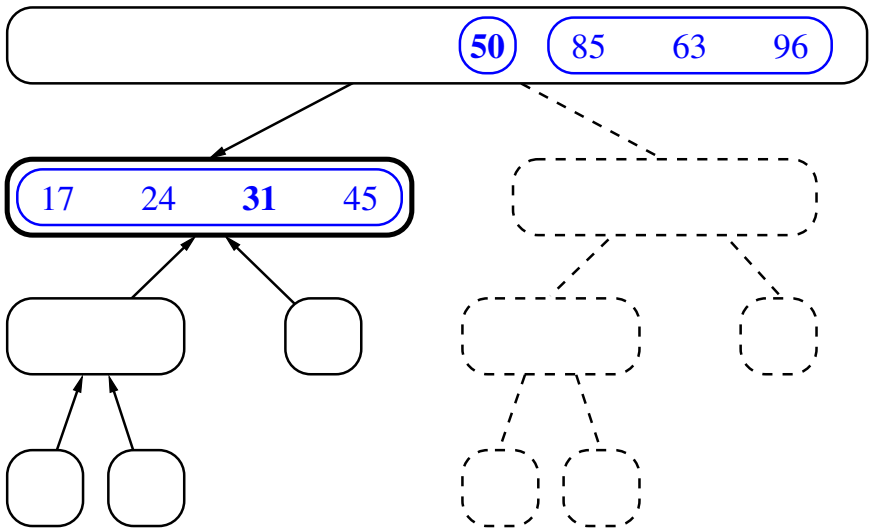

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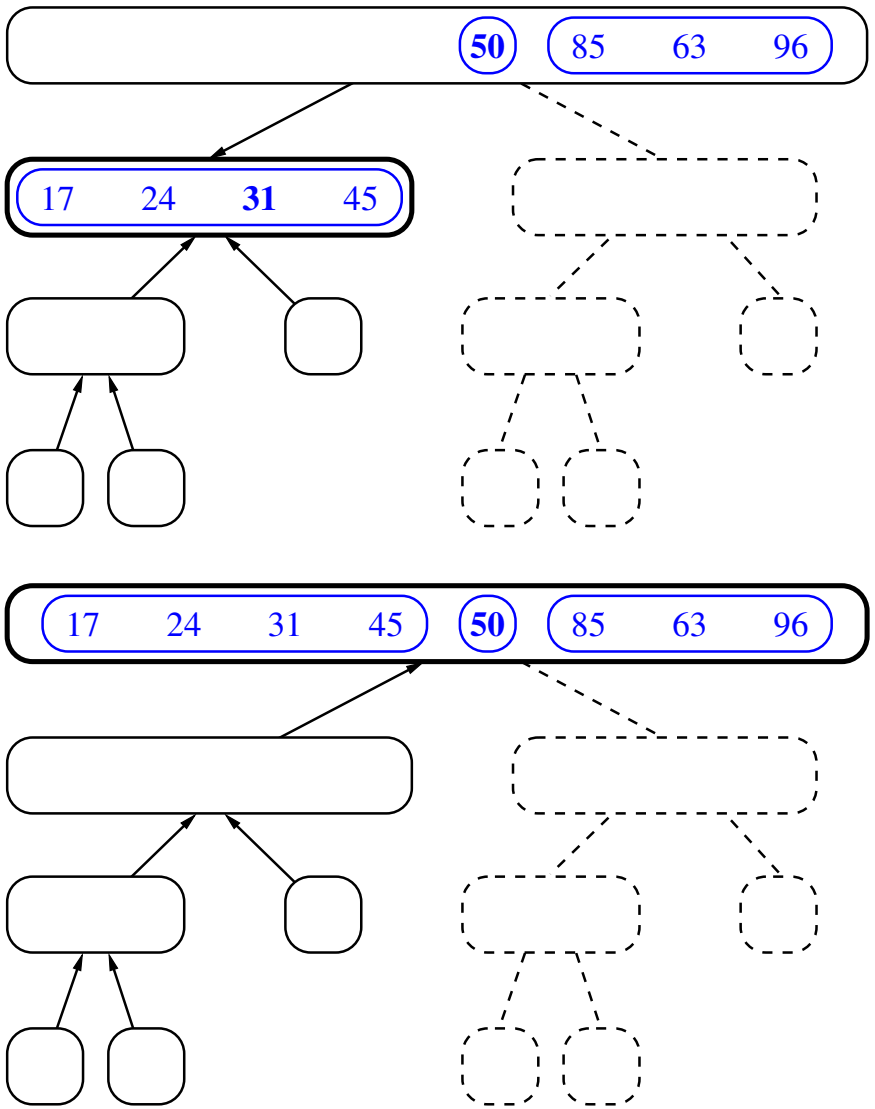

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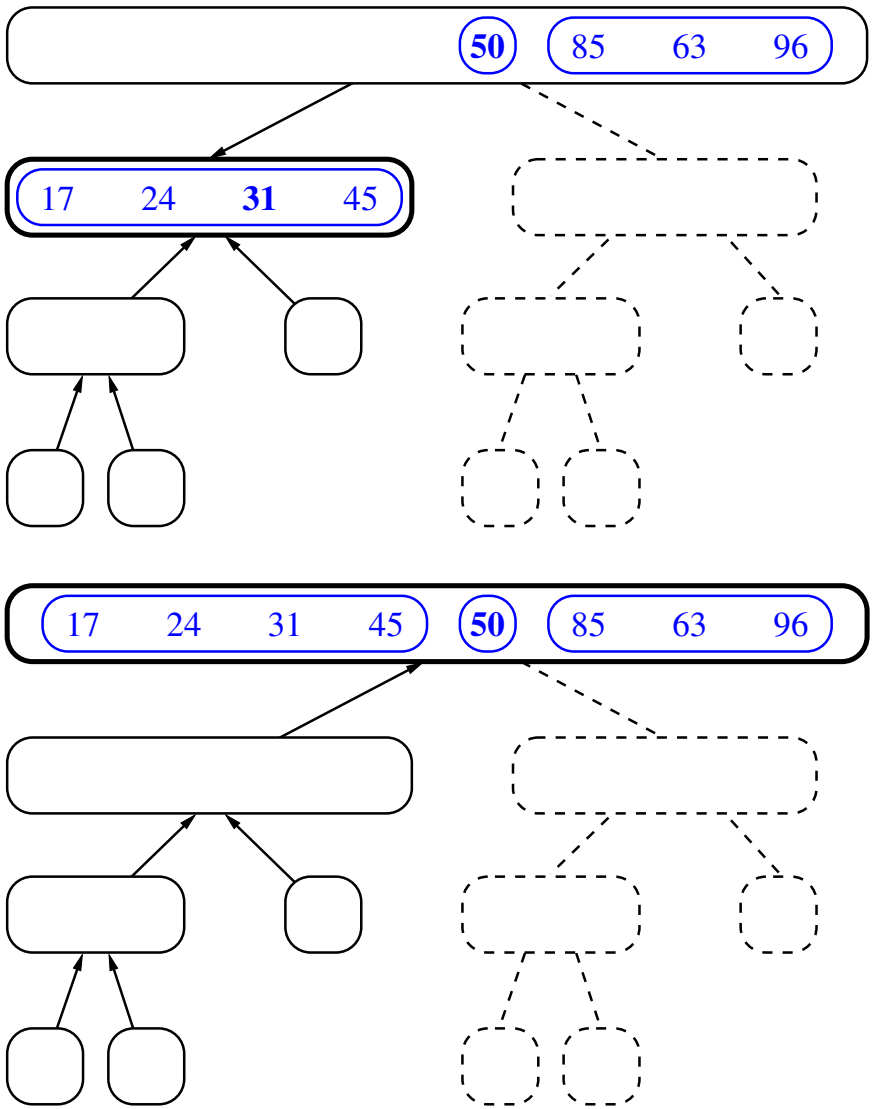

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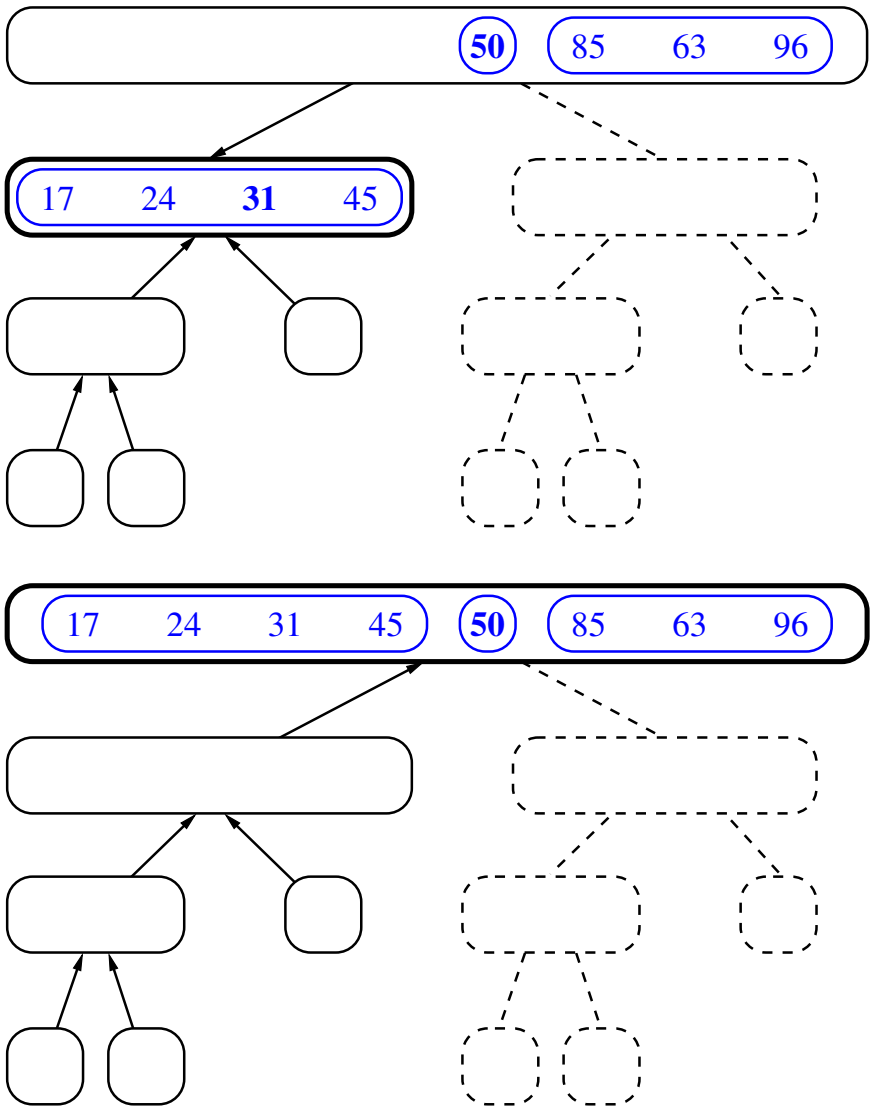

Implementation on Queue (bad one)

Algorithm 1: quickSortBad(Queue Q)

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4 (L, E, G)=partition(Q);
5 quickSort(L, comp);
6 quickSort(G, comp);
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Implementation on Queue (bad one)

Algorithm 2: quickSortBad(Queue Q)

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1 if Q.size<2 then
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Implementation on Queue (bad one)

Algorithm 3: quickSortBad(Queue Q)

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1 if Q.size<2 then
2 |   return;
3 end
4 (L, E, G)=partition(Q);
5 quickSort(L, comp);
6 quickSort(G, comp);
7 while (!L.isEmpty()) Q.enqueue(L.dequeue());
8 while (!E.isEmpty()) Q.enqueue(E.dequeue());
9 while (!G.isEmpty()) Q.enqueue(G.dequeue());
```

Queue partition(bad one)

Algorithm 4: (L,E,G) partition(Q)

```
1  pivot =Q.first();
2  L, E, G = new LinkedQueue();
3  while Q not empty do
4  |   nextElement = Q.dequeue();
5  |   switch compare nextElement with pivot do
6  |   |   case nextElement < pivot do
7  |   |   |   L.enqueue(nextElement);
8  |   |   end
9  |   |   case nextElement = pivot do
10 |   |   |   E.enqueue(nextElement);
11 |   |   end
12 |   |   case nextElement > pivot do
13 |   |   |   G.enqueue(nextElement);
14 |   |   end
15 |   end
16 end
```

Performance of partition (queue based)

- Time complexity:
  - remove each element y from Q
  - insert y into L, E or G, depending on the comparison with the pivot x
  - Each insertion and removal is at the beginning or at the end of a sequence, and hence takes $O(1)$ time
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  - Thus, the partition step of quick-sort takes $O(n)$ time
- Space complexity for partition:
  - $O(n)$

Implementation on Queue (bad one) in Java

```
public static <K> void quickSort(Queue<K> S, Comparator<K> comp) {
  int n = S.size();
  if (n < 2) return;
  K pivot = S.first();
  Queue<K> L = new LinkedQueue<>();
  Queue<K> E = new LinkedQueue<>();
  Queue<K> G = new LinkedQueue<>();
  while (!S.isEmpty()) {
    K element = S.dequeue();
    int c = comp.compare(element, pivot);
    if (c < 0) L.enqueue(element);
    else if (c == 0) E.enqueue(element);
    else G.enqueue(element);
  }
  quickSort(L, comp);
  quickSort(G, comp);
  while (!L.isEmpty())       S.enqueue(L.dequeue());
  while (!E.isEmpty())       S.enqueue(E.dequeue());
  while (!G.isEmpty())       S.enqueue(G.dequeue());
}
```

Quick sort on Array

- A better quickSort that does not need an extra array

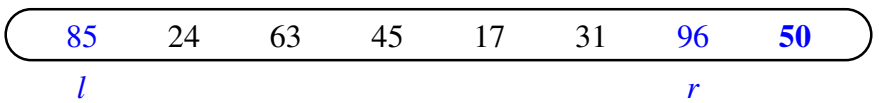
Algorithm 5: quickSortInPlace(A, int low, int high)

```
1 if low ≥ high then
2 |   return;
3 end
4 int p=partition(A, low, high);
5 quickSort(A, low, p-1);
6 quickSort(A, p+1, high);
```

In-place Partition (rewritten from the book)

Algorithm 6: partition(A, l, r)

```
1  pivot=A[r];
2  p=r;
3  r=r-1;
4  while l ≤ r do
5  |   while l ≤ r and A[l] < pivot do
6  |   |   l++;
7  |   end
8  |   while l ≤ r and A[r] > pivot do
9  |   |   r- -;
10 |   end
11 |   if l ≤ r then
12 |   |   swap(A[l], A[r]);
13 |   |   l++; r- -;
14 |   end
15 end
16 swap(A[l], A[p]);
17 return l;
```



In-place Partition (rewritten from the book)

Algorithm 7: partition(A, l, r)

```
1  pivot=A[r];
2  p=r;
3  r=r-1;
4  while l ≤ r do
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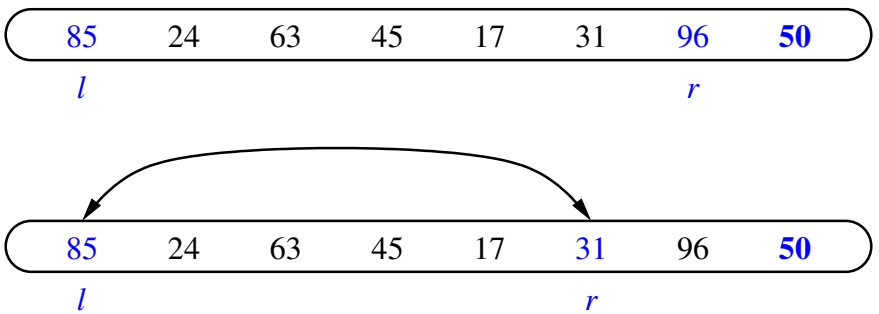

In-place Partition (rewritten from the book)

Algorithm 8: partition(A, l, r)

```
1  pivot=A[r];
2  p=r;
3  r=r-1;
4  while l ≤ r do
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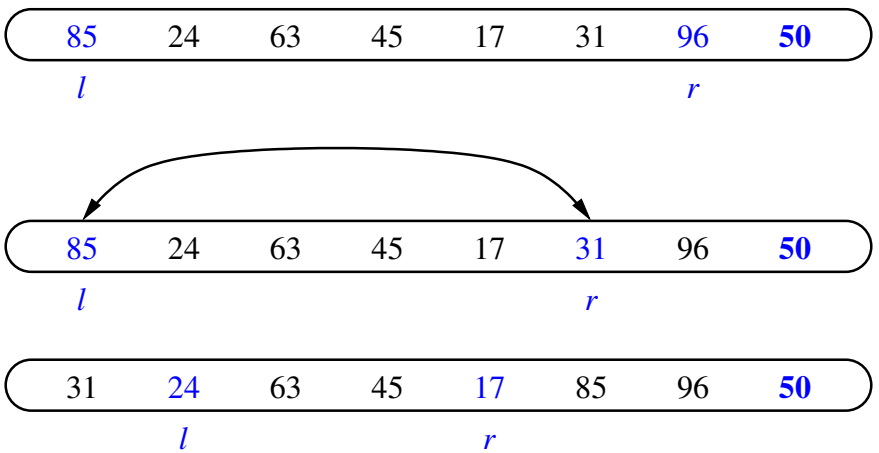

In-place Partition (rewritten from the book)

Algorithm 9: partition(A, l, r)

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2  p=r;
3  r=r-1;
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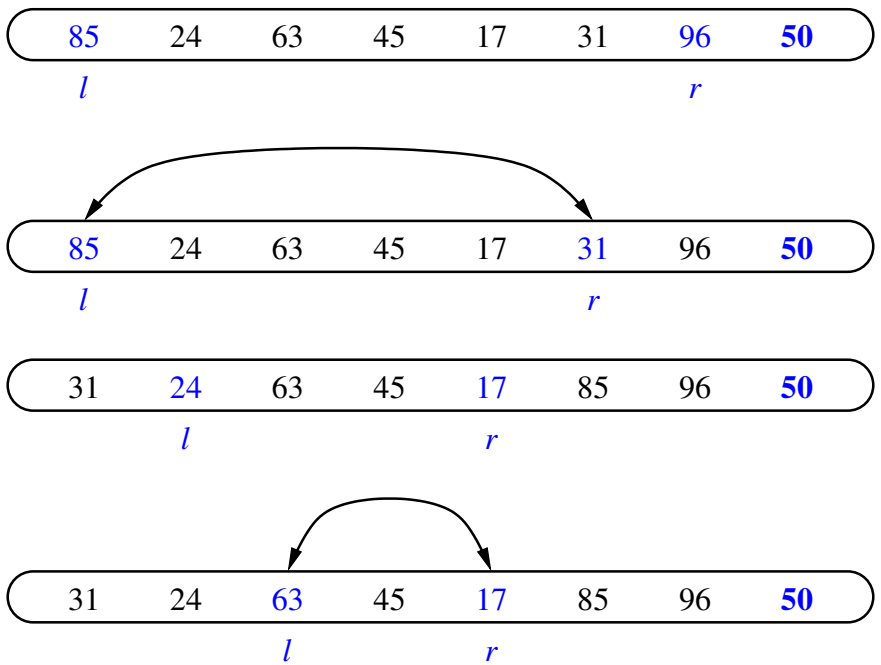
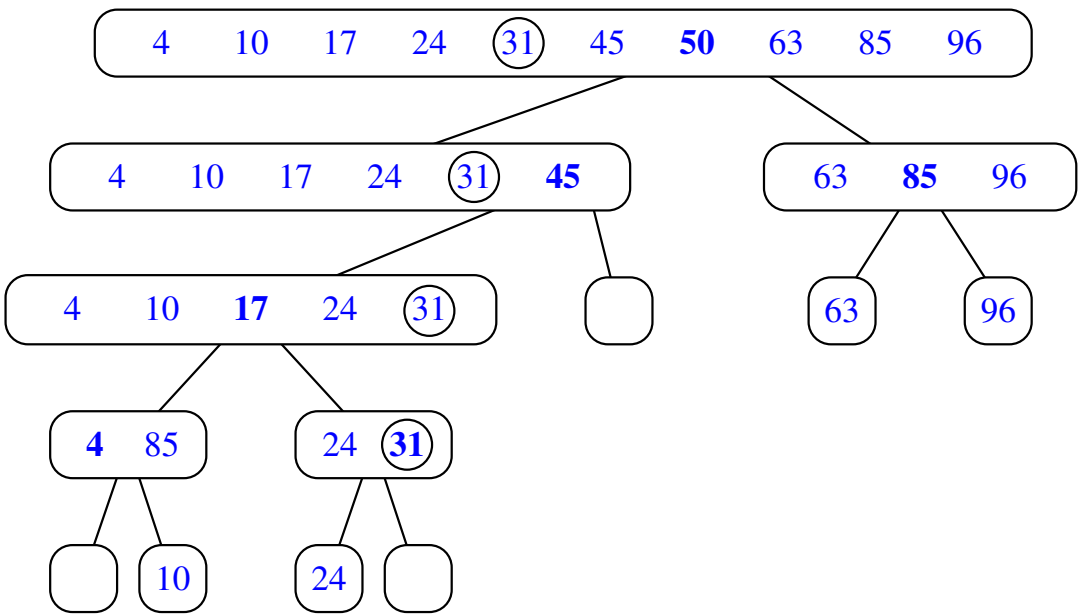


In-place sorting in Java

```
<K> void quickSortInPlace(K[]S,Comparator<K>comp,int a,int b){
    if (a >= b) return;
    int left = a;
    int right = b-1;
    K pivot = S[b];
    K temp;
    while (left <= right) {
      while (left<=right && comp.compare(S[left], pivot)<0) left++;
      while (left<=right && comp.compare(S[right], pivot)>0)
          right--;
      if (left <= right) {
        temp=S[left]; S[left]=S[right]; S[right]=temp;
        left++; right--;
      }
    }
    temp=S[left]; S[left]=S[b]; S[b]=temp;
    quickSortInPlace(S, comp, a, left - 1);
    quickSortInPlace(S, comp, left + 1, b);
  }
```

- An execution of quick-sort is depicted by a binary tree
- Each node represents a recursive call of quick-sort and stores
- Unsorted sequence before the execution and its pivot
- Sorted sequence at the end of the execution
- The root is the initial call
- The leaves are calls on subsequences of size 0 or 1



Worst-case for Quick sort

- The worst case for quick-sort occurs when the input is sorted
  - One of L and G has size n - 1 and the other has size 0
- The running time is proportional to the sum of:

$$n + (n-1) + \cdots + 2 + 1 = n(n+1)/2 \tag{1}$$

- Thus, the worst-case running time of quick-sort is $O(n^2)$
- Can avoid this be randomize the pivot

Average-case for Quick sort

- Average case number of comparisons is $1.39N \log N$.
- Probabilistic guarantee against worst case
- Better than MergeSort because
  - less space is needed
  - cache locality

Insertion Sort

Algorithm 10: Insertion-Sort(A)

```
1 for j = 2 to A.length do
2 |   key = A[j]
3 |   i = j-1
4 |   while i > 0 and A[i] > key do
5 |   |   A[i+1] = A[i]
6 |   |   i = i-1
7 |   end
8 |   A[i+1] = key
9 end
```

6 5 3 1 8 7 2 4

Best case running time is $O(n)$

- Let's focus on the number of comparisons.
- Checking number of all the 'operations' is more complicated.
- The best case is when A is already sorted. i.e.,

$$A[j-1] \leq A[j], \qquad (\text{for } j = 2, 3, \ldots, n)$$

- As this ensures that the condition of the *While* statement,

$$A[i] > key$$

  is false for $j = 2, 3, \ldots, n$,
- Inside the while loop there is only one comparison.
- The for loop checks n-1 elements

Thus, the number of comparison is

$$\sum_{j=2}^{n} 1 = n - 1.$$

Worst case number of comparisons

- In the worst case, the While condition will be tested for every possible value $i = j-1, j-2, \ldots, 1, 0$
- The cost for each while loop changes with j
- In total, it is

$$\sum_{j=2}^{n-1} j = \frac{n(n-1)}{2} - 1.$$

Average case number of comparisons

- For an "average" list, suppose that inside each while loop, we need to do half of the comparisons (and swaps),
- Hence the total number of comparisons is

$$\sum_{j=2}^{n-1} \sum_{j=2}^{n} j/2 = \frac{n(n-1)}{4} - 1.$$

- So the average running time of Insertion sort is $\Theta(N^2)$
- Any sorting algorithm that only swaps adjacent elements requires $\Omega(N^2)$ time because each swap removes only one inversion.
- If we are going to do better than $O(N^2)$, we are going to have to fix more than one inversion at a time
- How can we fix more than one inversion?
- Move the elements far away with each swap

Shell sort

- Named after Donald Shell – inventor of the algorithm
- Running time is $O(N^x)$, where x = 6/5, 5/4, 4/3, . . . , or 2, depending on "increment sequence"
- Shell sort uses repeated insertion sorts on selected subarrays of the larger array being sorted
- Multiple passes with changing subarrays

Shell Sort Example

Input	4	3	2	1

Shell Sort Example

Input	4	3	2	1
Two arrays with spacing=2	4	3	2	1

Shell Sort Example

Input	4	3	2	1
Two arrays with spacing=2	4	3	2	1
Sort the red array	2	3	4	1

Shell Sort Example

Input	4	3	2	1
Two arrays with spacing=2	4	3	2	1
Sort the red array	2	3	4	1
Sort the black array	2	1	4	3

Shell Sort Example

Input	4	3	2	1
Two arrays with spacing=2	4	3	2	1
Sort the red array	2	3	4	1
Sort the black array	2	1	4	3
Reduce the spacing =1	2	1	4	3
inverse 2/1	1	2	4	3

Shell Sort Example

Input	4	3	2	1
Two arrays with spacing=2	4	3	2	1
Sort the red array	2	3	4	1
Sort the black array	2	1	4	3
Reduce the spacing =1	2	1	4	3
inverse 2/1	1	2	4	3
inverse 4/3	1	2	4	3

Shell Sort Example

Input	4	3	2	1
Two arrays with spacing=2	4	3	2	1
Sort the red array	2	3	4	1
Sort the black array	2	1	4	3
Reduce the spacing =1	2	1	4	3
inverse 2/1	1	2	4	3
inverse 4/3	1	2	4	3

- number of inversions in shell sort: 4
- number of inversions in insertion sort: 1+2+3=6

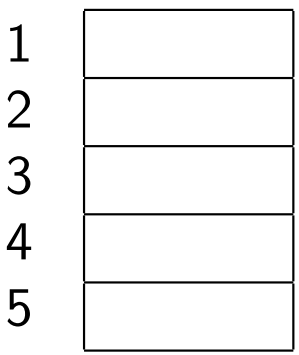
Intuition of Shell Sort

- Insertion sort runs fast on nearly sorted sequences
- Immediate termination when proper spot is found
- Do several passes of Insertion sort on different subsequences of elements
- The subsequences stay sorted from pass to pass

BucketSort, BinSort, CountingSort

Motivating example: If all values to be sorted are known to be between 1 and B

- create an array count of size B,
- increment counts while traversing the input,
- and finally output the result.
- e.g., Input=(4, 5, 1, 3, 5)
- B is 5. create 5 buckets using array:



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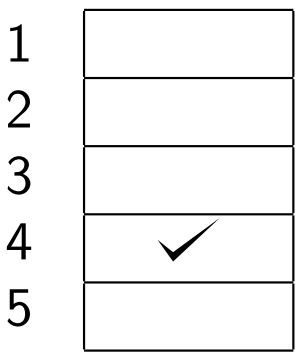

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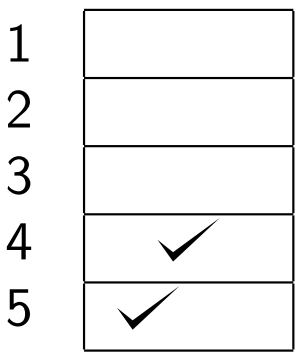

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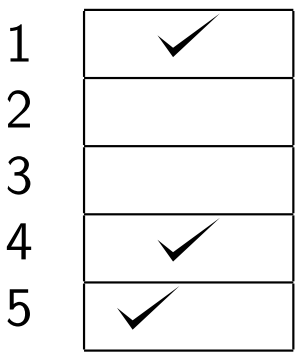

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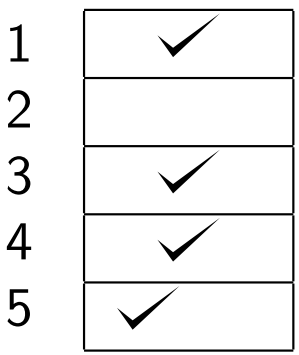

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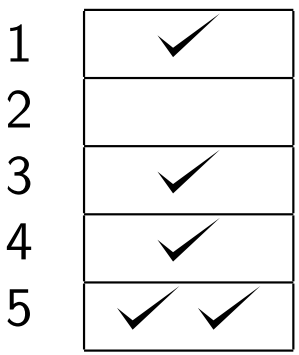

- We can extend the idea to sort real numbers
- Bucket can contain distinct items

Time Complexity

- What is the complexity?

Time Complexity

- What is the complexity? $O(n + B)$
- Depending on B
  - What if...
  - B is a constant
  - B is proportional to n
  - B is a very large constant (2^{32})

Fixing impracticality: RadixSort

- Radix Sort:generalization of Bucket Sort for large integer keys.
- Origins go back to the 1890 census.
- Radix= "The base of a number system"
- We use 10 for convenience, but could be anything
- Ideas:
  - Bucket Sort on one digit at a time
  - After k-th sort, the last k digits are sorted
  - Set number of buckets: B = radix.

First pass

- Input: 478, 537, 9, 721, 3, 38, 123, 67

	0	1	2	3	4	5	6	7	8	9
Sort on 1's										

First pass

- Input: 478, 537, 9, 721, 3, 38, 123, 67

	0	1	2	3	4	5	6	7	8	9
Sort on 1's									478	

First pass

- Input: 478, 537, 9, 721, 3, 38, 123, 67

	0	1	2	3	4	5	6	7	8	9
Sort on 1's								537	478	

First pass

- Input: 478, 537, 9, 721, 3, 38, 123, 67

	0	1	2	3	4	5	6	7	8	9
Sort on 1's								537	478	9

First pass

- Input: 478, 537, 9, 721, 3, 38, 123, 67

	0	1	2	3	4	5	6	7	8	9
Sort on 1's		721						537	478	9

First pass

- Input: 478, 537, 9, 721, 3, 38, 123, 67

	0	1	2	3	4	5	6	7	8	9
Sort on 1's		721		3				537	478	9

First pass

- Input: 478, 537, 9, 721, 3, 38, 123, 67

	0	1	2	3	4	5	6	7	8	9
Sort on 1's		721		3				537	478 38	9

First pass

- Input: 478, 537, 9, 721, 3, 38, 123, 67

	0	1	2	3	4	5	6	7	8	9
Sort on 1's		721		3 123				537	478 38	9

First pass

- Input: 478, 537, 9, 721, 3, 38, 123, 67

	0	1	2	3	4	5	6	7	8	9
Sort on 1's		721		3 123				537 67	478 38	9

First pass

- Input: 478, 537, 9, 721, 3, 38, 123, 67

	0	1	2	3	4	5	6	7	8	9
Sort on 1's		721		3 123				537 67	478 38	9

- Output of first pass: 721, 3, 123, 537, 67, 478, 38, 9
- used as the input of second pass

Second pass

- Input: 721, 3, 123, 537, 67, 478, 38, 9

	0	1	2	3	4	5	6	7	8	9
Sort on 10's										

Second pass

- Input: 721, 3, 123, 537, 67, 478, 38, 9

	0	1	2	3	4	5	6	7	8	9
Sort on 10's			721							

Second pass

- Input: 721, 3, 123, 537, 67, 478, 38, 9

	0	1	2	3	4	5	6	7	8	9
Sort on 10's	03		721							

Second pass

- Input: 721, 3, 123, 537, 67, 478, 38, 9

	0	1	2	3	4	5	6	7	8	9
Sort on 10's	03		721 123							

Second pass

- Input: 721, 3, 123, 537, 67, 478, 38, 9

	0	1	2	3	4	5	6	7	8	9
Sort on 10's	03		721 123	537						

Second pass

- Input: 721, 3, 123, 537, 67, 478, 38, 9

	0	1	2	3	4	5	6	7	8	9
Sort on 10's	03		721 123	537			67			

Second pass

- Input: 721, 3, 123, 537, 67, 478, 38, 9

	0	1	2	3	4	5	6	7	8	9
Sort on 10's	03		721 123	537			67	478		

Second pass

- Input: 721, 3, 123, 537, 67, 478, 38, 9

	0	1	2	3	4	5	6	7	8	9
Sort on 10's	03		721 123	537 38			67	478		

Second pass

- Input: 721, 3, 123, 537, 67, 478, 38, 9

	0	1	2	3	4	5	6	7	8	9
Sort on 10's	03 09		721 123	537 38			67	478		

Second pass

- Input: 721, 3, 123, 537, 67, 478, 38, 9

	0	1	2	3	4	5	6	7	8	9
Sort on 10's	03 09		721 123	537 38			67	478		

- Output of second pass: 3,9,721, 123, 537, 38, 67, 478
- used as the input of second pass

Third pass

- Input: 3,9,721, 123, 537, 38, 67, 478

	0	1	2	3	4	5	6	7	8	9
Sort on 100's										

- Output: 3, 9, 38, 67, 123, 537, 721.

Third pass

- Input: 3,9,721, 123, 537, 38, 67, 478

	0	1	2	3	4	5	6	7	8	9
Sort on 100's	003									

- Output: 3, 9, 38, 67, 123, 537, 721.

Third pass

- Input: 3,9,721, 123, 537, 38, 67, 478

	0	1	2	3	4	5	6	7	8	9
Sort on 100's	003 009									

- Output: 3, 9, 38, 67, 123, 537, 721.

Third pass

- Input: 3,9,721, 123, 537, 38, 67, 478

	0	1	2	3	4	5	6	7	8	9
Sort on 100's	003 009							721		

- Output: 3, 9, 38, 67, 123, 537, 721.

Third pass

- Input: 3,9,721, 123, 537, 38, 67, 478

	0	1	2	3	4	5	6	7	8	9
Sort on 100's	003 009	123						721		

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	0	1	2	3	4	5	6	7	8	9
Sort on 100's	003 009	123				537		721		

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- Input: 3,9,721, 123, 537, 38, 67, 478

	0	1	2	3	4	5	6	7	8	9
Sort on 100's	003 009 038	123				537		721		

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Third pass

- Input: 3,9,721, 123, 537, 38, 67, 478

	0	1	2	3	4	5	6	7	8	9
Sort on 100's	003 009 038 067	123				537		721		

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	0	1	2	3	4	5	6	7	8	9
Sort on 100's	003 009 038 067	123			478	537		721		

- Output: 3, 9, 38, 67, 123, 537, 721.

Complexity of Radix sort

- Input size, n
- In our example,
  - Number of buckets, B $= 10$
  - Maximum value, three digits, $M < 10^3$
  - Number of passes, $P = \log_B M = 3$
- Run time proportional to

$$(B + n)P = (B + n) \log_B M \tag{2}$$

- In practice, B should be much larger than 10, so that P is a small number.

Comparison of sorting algorithms

	Method	Time Best	Worst	Avg	In-place	Stable
Comparison-based	Selection	n^2	n^2	n^2	$O(1)$	No
	Insertion	n	n^2	n^2	$O(1)$	Yes
	Shell			$n^{6/5}$	$O(1)$	No
	HeapSort	$n \log n$	$n \log n$	$n \log n$	$O(1)$	No
	Merge	$n \log n$	$n \log n$	$n \log n$	$O(n)$	Yes
	Quick	$n \log n$	n^2	$n \log n$	$O(1)$.	No
Counting-based	Bucket			O(n+B)	O(B)	Yes
	Radix			O(n+B)	O(B)	Yes

- Quick sort is often used in practice. (e.g., in Java)
  - It is in-place sorting. Save space compared with mergeSort.
- merge Sort: good for external memory

Takeaways

- Evaluation of sorting algorithms: Time, space, stability, data dependency, application dependency
- Quick sort. Divide and conquer. Time complexity. Worst case and average case. Its comparison with merge sort.
- Shell sort.
- Bucket and Radix sort.
- Readings: Goodrich et al. P532-562.
- YouTube video for 15 sorting algorithms in 6 minutes